

MECH 632

Intermediate Spaceflight Dynamics

Attitude Dynamics

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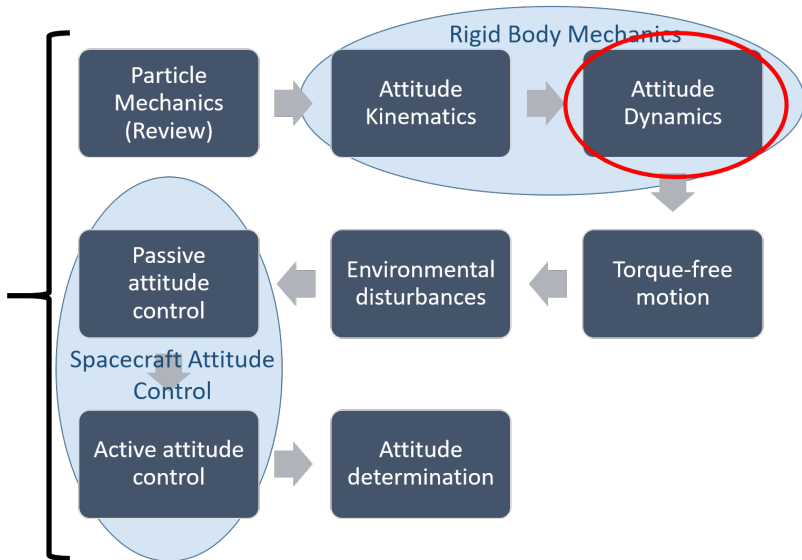
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Where are we?



Spacecraft Attitude Determination and Control





- 1 Lesson Goals
- 2 Rigid Body Properties
- 3 Rigid Body Dynamics
- 4 The complete Rigid Body EOMs (Kinematics + Dynamics)
- 5 Lesson Summary



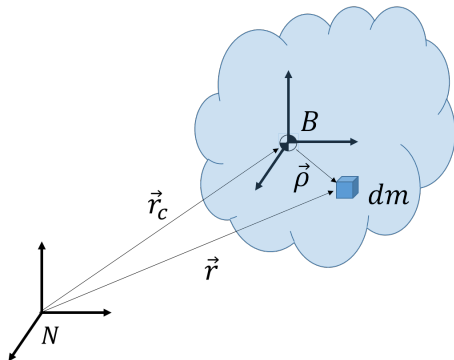
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- ▶ Define the angular momentum, moment of inertia, and kinetic energy of a rigid body
- ▶ Identify important properties of the rigid body's Moment of Inertia matrix
- ▶ Starting from Newton's Second Law, rigorously derive the dynamic equations for a rigid body



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Fundamental Dynamics Equation:

The total moment (torque) acting on the body, τ , about the center of mass (c.o.m.) is related to the total angular momentum about the c.o.m.:

$$\boxed{\dot{\vec{H}} = \vec{\tau}} \quad (1)$$

Conservation of Angular Momentum:

If no external torques are acting on the body, angular momentum is conserved:

$$\dot{\vec{H}} = 0$$

The total angular momentum of the rigid body is:

$$\vec{H} = \int_B \vec{\rho} \times \dot{\vec{\rho}} dm,$$

and

$$\dot{\vec{\rho}} = \frac{{}^N d}{{}^N dt} \vec{\rho} = \frac{{}^B d}{{}^B dt} \vec{\rho} + \vec{\omega}_{BN} \times \vec{\rho}$$

Then, the angular momentum of the rigid body is:

$$\vec{H} = \int_B \vec{\rho} \times (\vec{\omega}_{BN} \times \vec{\rho}) dm = \left(\int_B -[\vec{\rho}^\times][\vec{\rho}^\times] dm \right) \vec{\omega}_{BN}$$

Moment of Inertia (MOI) Matrix

Resolve all quantities in the body-fixed CCS:

$$\boldsymbol{\rho} = [x, y, z]^T, \quad \boldsymbol{\omega}_{BN} = [\omega_1, \omega_2, \omega_3]^T$$

$$\mathbf{H}^B = \int_B \begin{bmatrix} y^2 + z^2 & -xy & -xz \\ -xy & x^2 + z^2 & -yz \\ -xz & -yz & x^2 + y^2 \end{bmatrix} dm \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix}$$

Therefore,

$$\boxed{\mathbf{H}^B = \mathbf{J} \boldsymbol{\omega}_{BN}} \quad (2)$$

- ▶ $\mathbf{J}$ is the *inertia matrix* of the rigid body in the body CCS
- ▶ Diagonal terms are called *moments* of inertia
- ▶ Off-diagonal terms are called *products* of inertia

Note: If the angular momentum is written in vector form (unresolved) then $\mathbf{J}$ is called the inertia *dyadic*



- ▶ Representation of a physical property of the rigid body (i.e., does not change in a body-fixed reference frame)
- ▶ All elements are real-valued
- ▶ J is symmetric and positive definite
 - ▶ What does this say about its eigenvalues?
- ▶ Rotation to a different CCS:

$$J^B = C_{BB'} J^{B'} C_{B'B} \quad \text{or} \quad J^{B'} = C_{B'B} J^B C_{BB'}$$

- ▶ The following triangle inequalities hold:

$$J_{11} + J_{22} \geq J_{33}$$

$$J_{11} + J_{33} \geq J_{22}$$

$$J_{22} + J_{33} \geq J_{11}$$

- ▶ For any rigid body there exists a set of axes for which the products of inertia are all zero (i.e. J is diagonal); these are the *principal axes* and the corresponding moments of inertia are the *principal moments of inertia*
- ▶ Can be found by solving the eigenvalue/eigenvector problem:

$$[\lambda I - J]e = 0$$

where I is a 3 identity matrix

- ▶ The principal moments are the eigenvalues given by the roots of the characteristic equation $\det(\lambda I - J) = 0$
- ▶ The three normalized eigenvectors $\hat{e}_{1,2,3}$ are the principal axes (expressed in the non-principal CCS), then:

$$C_{B'B} = \begin{bmatrix} \hat{e}_1 \\ \hat{e}_2 \\ \hat{e}_3 \end{bmatrix}, \text{ and } J^{B'} = C_{B'B} J^B C_{B'B}^T$$



Define the principal MOI as J_x , J_y , and J_z :

- ▶ A rigid body with $J_x = J_y = J_z$ is called *spherically symmetric*
- ▶ A rigid body with $J_x = J_y$, $J_x = J_z$, or $J_y = J_z$ is called *axially symmetric*
- ▶ A rigid body with $J_x \neq J_y \neq J_z$ is called *triaxial* or *asymmetric*



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Euler's Rotational Equations of Motion



According to Newton's Second Law, the rotational motion of a rigid body about its center of mass is given as:

$$\dot{\vec{H}} = \vec{\tau},$$

Taking the time derivative:

$$\begin{aligned}\dot{\vec{H}} &= \frac{{}^N d}{dt} \vec{H} = \frac{{}^B d}{dt} \vec{H} + \vec{\omega}_{BN} \times \vec{H} \\ \frac{{}^B d}{dt} \vec{H} &= \frac{{}^B d}{dt} (J \vec{\omega}_{BN})\end{aligned}$$

Note that $\frac{{}^B d}{dt} J = 0$ and $\frac{{}^B d}{dt} \vec{\omega}_{BN} = \frac{{}^N d}{dt} \vec{\omega}_{BN} = \dot{\vec{\omega}}_{BN}$.

Resolving all quantities in the body-fixed CCS:

$$\boxed{J \dot{\vec{\omega}}_{BN} + [\vec{\omega}_{BN}^\times] J \vec{\omega}_{BN} = \vec{\tau}} \quad (3)$$



Choosing a body-fixed CCS aligned with the principal axes:

$$J_x \dot{\omega}_x + (J_z - J_y) \omega_z \omega_y = \tau_x \quad (4)$$

$$J_y \dot{\omega}_y + (J_x - J_z) \omega_x \omega_z = \tau_y \quad (5)$$

$$J_z \dot{\omega}_z + (J_y - J_x) \omega_y \omega_x = \tau_z \quad (6)$$

These equations are:

- ▶ First-order ODE's
- ▶ Nonlinear and coupled
- ▶ Solved numerically (most of the time)



- ▶ Angular momentum has been defined as

$$\mathbf{H} = J\boldsymbol{\omega}_{BN}$$

- ▶ The rotational Kinetic Energy is defined as

$$T = \frac{1}{2}\boldsymbol{\omega}_{BN}J\boldsymbol{\omega}_{BN}$$

In a principal body-fixed CCS the following conditions are true:

$$\begin{aligned} ||\mathbf{H}||^2 &= (J_x\omega_x)^2 + (J_y\omega_y)^2 + (J_z\omega_z)^2 \\ 2T &= J_x\omega_x^2 + J_y\omega_y^2 + J_z\omega_z^2 \end{aligned}$$



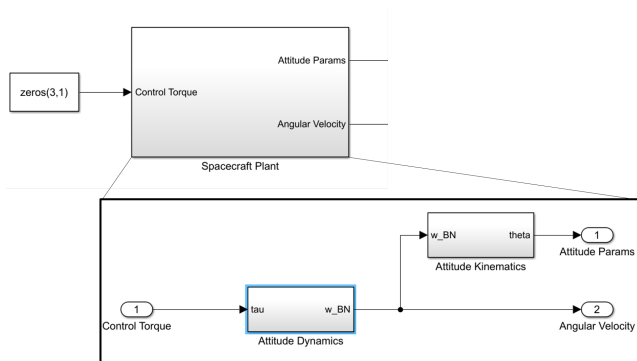
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Putting it all together...

Combining the Kinematic and Dynamic Equations gives the complete description of the rotational rigid body motion:

$$\dot{\bar{q}} = \frac{1}{2} \begin{bmatrix} [\bar{q}^\times] + q_4 I \\ -\bar{q}^T \end{bmatrix} \omega_{BN} \quad (7)$$

$$J\dot{\omega}_{BN} = \tau - [\omega_{BN}^\times] J \omega_{BN} \quad (8)$$





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- ▶ We have derived equations for the dynamics of a rigid body
- ▶ Given initial conditions of a spacecraft, we can now model how its attitude and angular velocity change over time
- ▶ The next step is to analyze, and eventually be able to control this motion